# Rajeev Jain


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Principal Specialist, Research Software Engineering | AI Systems | HPC | Scientific Computing

At Argonne National Laboratory **since 2009** | **2× R&D 100 Awards** (CANDLE 2023, FLASH-X 2022) | Lead developer, **UXarray** (DOE labs & NCAR)

## Professional Profile

Principal research software engineer, at Argonne since 2009, delivering production systems at the intersection of AI infrastructure, scientific computing, and HPC. Led UXarray from zero to a Python library used at DOE labs and NCAR, including an MCP server that lets AI agents inspect, analyze, and visualize production Earth-system meshes on HPC clusters via Globus Compute  no SSH, no scripts, full provenance. Pairs that with a verification habit: traced silent numerical drift on a production mesh to catastrophic cancellation and fixed it upstream, and recovered a hidden 1.67× slowdown in a faithful C++-to-Python port. Ran 10,000+ ML training experiments using Swift/T across Summit, Theta, and Cori for cancer drug response research; applied noise injection and counterfactual analysis to identify genes driving tumor classification decisions. Reduced FLASH-X checkpoint overhead 40–70% on Summit via async HDF5 and lossy compression. Generated a 101M-element MONJU reactor mesh on 712 processors in about 7 minutes  a mesh the serial path could not build at all. Two R&D 100 Awards (2022, 2023). Work spans Python platforms and MCP server development, C++ and Fortran HPC codebases, exascale GPU porting (Intel XPU / CUDA), and cross-institution program coordination.

## Education

**University of Chicago** — 2020
M.S. in Computer Science

**Arizona State University** — 2009
M.S. in Structural Engineering
Advisors: Dr. S.D. Rajan (ASU), Dr. Ashok Belegundu (PSU)
Graduate Fellowship Recipient (2007–2009)

**Indian Institute of Technology (ISM) Dhanbad** — 2006
B.Tech. in Mechanical Engineering

## Professional Experience

**Argonne National Laboratory** — **2009 – Present**
*Principal Specialist, Research Software Engineering* — *Mathematics and Computer Science Division, Chicago, IL*

- Principal software engineer and program lead across six multi-year DOE research software programs spanning climate analytics, scientific AI, simulation infrastructure, and exascale HPC porting  from initial architecture through production release, CI/CD, and cross-institution coordination.

**University of Chicago** — **2023 – Present**
*Staff At-Large* — *Consortium for Advanced Science and Engineering (CASE), Chicago, IL*

- Joint appointment supporting cancer pharmacogenomics and Earth science collaborations.
- Mentor students and research staff on software design, experimentation, and reproducibility.

**Arizona State University** — **2007 – 2009**
*Research and Teaching Assistant* — *Structural and Computational Mechanics Lab, Tempe, AZ*

- Conducted FEM-based research in blast mitigation and supported structural analysis coursework.

**Earlier Industry Experience** — **2005 – 2007**
*Project Engineer, Wipro Technologies; Engineering Intern, Tata Motors* — *India*

## Research Software Leadership and Projects

**UXarray** **2022 – Present**
*Lead Developer* *Open-source climate analysis platform*

- Lead developer for a Python library enabling analysis of native unstructured climate and weather grids, used by NCAR, DOE labs, and universities.
- Designed the core API, conservative zonal averaging via exact spherical polygon intersection, vector calculus operators (gradient, divergence, curl), YAC remapping backend, multi-grid support, geometry operators, grid I/O for ESMF / MPAS / SCRIP / HEALPix, automated tests, CI/CD, and release workflows.
- Built the UXarray MCP server: 20+ typed tools allowing AI agents to inspect, subset, visualize, and analyze production Earth-system meshes locally or on HPC clusters (Argonne Improv, UCAR Derecho) via Globus Compute, with structured provenance on every operation. Presented at SciFoundationModels 2026 (SciFM26).
- Traced 0.68 m of positional drift on a production mesh to catastrophic cancellation in a spherical cross product and replaced the kernel with error-free-transformation arithmetic (`two_prod`), merged as UXarray PR #1513; recovered a 1.67× speedup in a Numba hot path by removing heap allocation introduced in a C++-to-Python translation.
- Tutorials and presentations at SC24, AMS 2024/2025/2026, SciPy 2023, EGU 2024/2025/2026, AGU, and ESDS.

**ncvis** **2026 – Present**
*Maintainer* *NetCDF visualizer for structured and unstructured grids (DOE SEATS)*

- Took over maintenance of the SEATS project's C++/wxWidgets visualizer in August 2026: reviewed and merged the community pull-request backlog (1D line plots, color-scale centering, macOS XQuartz refresh, CMake link order), building each branch on both build paths and running it against generated netCDF fixtures before merging.
- Diagnosed a macOS link failure caused by `target_link_options` de-duplicating repeated `-framework` arguments and replaced the `wx-config` parsing with `find_package` results; restored CMake 4 compatibility; fixed the `-var` startup option.

**SFNO Climate Emulator on Aurora** **2025 – 2026**
*Software Engineer* *AI climate emulation on exascale Intel GPUs*

- Ported the SFNO climate emulator to Intel XPU on Aurora: PMIX/PALS launcher mapping, XPU/CUDA device branching, device-aware AMP (gradient scaling on CUDA, bf16 on XPU), and first stable DDP baseline on Aurora.
- Established smoke-test and queue-aware bring-up scripts; separated launcher, backend, and device-placement failure modes from sharding bugs.

**CANDLE / IMPROVE** **2017 – 2025**
*Core Contributor* *Cancer AI infrastructure and benchmarking*

- Built the CANDLE/Supervisor HPO workflow using Swift/T, mlrMBO (Bayesian/GP), DEAP (genetic algorithms), and Hyperopt/TPE; ran 10,000+ model training experiments across Summit, Theta, and Cori at up to 1,000 concurrent jobs.
- Applied noise injection and counterfactual analysis to the NT3 tumor classification benchmark; identified PLOD2, RGS5, TP53I13, and related genes as decision-boundary drivers confirming the model used biologically meaningful signal.
- Led the IMPROVE benchmark framework: `improvelib` Python package with standardized loaders, canonical train/test splits, and evaluation metrics; GitHub Actions CI/CD covering unit tests, end-to-end Docker pipelines, and parallel cross-study analysis (CSA) via Parsl.
- Designed and implemented UNO, a dual-branch neural network (separate drug and cell-line embeddings) that participates in the IMPROVE CSA benchmark; results published in *Briefings in Bioinformatics* (2026).
- Supported a 15+ researcher multi-lab collaboration. R&D 100 Award, 2023.

**FLASH-X** **2019 – 2024**
*I/O and Compression Lead* *Multiphysics simulation software*

- Implemented asynchronous HDF5 I/O with Argobots and integrated SZ3 / ZFP compression for checkpoint and restart workflows.
- Reduced checkpoint overhead by 40-70% on Summit and achieved storage savings above 50%.
- Enabled cross-checkpoint restart between AMReX and Paramesh solvers. Work contributed to an R&D 100 Award in 2022.

**MeshKit** **2009 – 2016**
*Principal Investigator / Software Lead* *DOE NEAMS reactor mesh generation*

- Principal investigator for MeshKit within DOE NEAMS, leading open-source C++ toolkit for automated nuclear reactor core mesh generation.

- Developed parallel CoreGen workflow: 712 processors, 101 million hexahedral elements, 14 GB MONJU reactor mesh in about 7 minutes serial path could not run at all.
- Built AssyGen/CoreGen pipeline encoding reactor lattice hierarchy; supported PostBL boundary-layer generation and multi-format mesh workflows.

**Urban ECP** **2016 – 2019**

*Software Lead* *DOE urban microclimate and CFD workflows*

- Led seed-funded urban exascale proposal with Charlie Catlett (Argonne / University of Chicago); developed a three-layer coupled simulation workflow: WRF/HRRR mesoscale weather → EnergyPlus building energy → Nek5000 wall-resolved LES for downtown Chicago.
- Built the full 3D urban geometry model for the CFD domain, including architecturally complex structures such as Lake Point Tower.
- Published in *Journal of Building Performance Simulation* (2020); presented at AGU 2018/2019 and International Conference on Urban Climate 2018.

## Funding and Research Programs

- **DOE SEATS** (active): Simplifying ESM Analysis Through Standards; UXarray and ncvis development.
- **NSF Raijin** (active): collaborative research in climate model analysis.
- **DOE ECP CANDLE** (2017 – 2023): core contributor on cancer AI infrastructure.
- **DOE NEAMS** (2009 – 2016): principal investigator for MeshKit and reactor mesh generation workflows.
- **Urban ECP** (2016 – 2019): software lead and coordinator for urban microclimate and CFD workflows across multiple labs.

## Awards & Recognition

- R&D 100 Award — CANDLE, Cancer Distributed Learning Environment (2023).
- R&D 100 Award — FLASH-X, multiphysics simulation software (2022).
- DOE SBIR Phase I & II — RGG (Reactor Geometry & mesh Generator) tool, awarded to Kitware Inc. (2014 – 2017).
- ATPESC Scholar, Argonne Training Program on Extreme-Scale Computing (2015).
- Best Paper Award, International Meshing Roundtable (2010).
- University Graduate Fellowship, Arizona State University (2007 – 2009).
- First Prize, Low-Budget Car Design Contest — Minda Ltd. & IIT Kharagpur (2004).

## Peer-Reviewed Journal Articles

2026 Chen, H., Ullrich, P. A., Panetta, J., Marsico, D., Hanke, M., **Jain, R.**, Zhang, C., et al. "Accurate and Robust Geometric Algorithms for Regridding on the Sphere." *EGUsphere* (preprint).

2026 Partin, A., Vasanthakumari, P., Narykov, O., Wilke, A., Koussa, N., Jones, S., Zhu, Y., **Jain, R.**, Overbeek, J. C., Fernando, G. D., Sanchez-Villalobos, C., Garcia-Cardona, C., Mohd-Yusof, J., Chia, N., Wozniak, J. M., Ghosh, S., Pal, R., Brettin, T. S., Weil, M. R., and Stevens, R. L. "Benchmarking community drug response prediction models: datasets, models, tools, and metrics for cross-dataset generalization analysis." *Briefings in Bioinformatics* 27(1), bbaf667.

2022 Dubey, A., Weide, K., O'Neal, J., Dhruv, A., Couch, S., Harris, J. A., Klosterman, T., **Jain, R.**, Rudi, J., Messer, B., Pajkos, M., Carlson, J., Chu, R., Wahib, M., Chawdhary, S., Ricker, P. M., Lee, D., Antypas, K., Riley, K. M., Daley, C., Ganapathy, M., Timmes, F. X., Townsley, D. M., Vanella, M., Bachan, J., Rich, P. M., Kumar, S., Endeve, E., Hix, W. R., Mezzacappa, A., and Papatheodore, T. "Flash-X: A multiphysics simulation software instrument." *SoftwareX*.

2022 Harris, J. A., Chu, R., Couch, S. M., Dubey, A., Endeve, E., Georgiadou, A., **Jain, R.**, Kasen, D., Laiu, M. P., Messer, O. E. B., O'Neal, J., Sandoval, M. A., and Weide, K. "Exascale models of stellar explosions: quintessential multi-physics simulation." *International Journal of High Performance Computing Applications*.

2020 **Jain, R.**, Luo, X., Sever, G., Hong, T., and Catlett, C. "Representation and evolution of urban weather boundary conditions in downtown Chicago." *Journal of Building Performance Simulation*.

2018 Wozniak, J. M., **Jain, R.**, Balaprakash, P., Ozik, J., Collier, N. T., Bauer, J., Xia, F., Brettin, T., Stevens, R., Mohd-Yusof, J., Garcia-Cardona, C., Van Essen, B., and Baughman, M. "CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research." *BMC Bioinformatics*.

2014 Mahadevan, V. S., Merzari, E., Tautges, T., **Jain, R.**, Obabko, A., Smith, M., and Fischer, P. "High-resolution coupled physics solvers for analysing fine-scale nuclear reactor design problems." *Philosophical Transactions of the Royal Society A*.

2012 Tautges, T. J. and **Jain, R.** "Creating geometry and mesh models for nuclear reactor core geometries using a lattice hierarchy-based approach." *Engineering with Computers*.

2008 Argod, V., Belegundu, A. D., Aziz, A., Agrawala, V., **Jain, R.**, and Rajan, S. D. "MPI-enabled shape optimization of panels subjected to air blast loading." *International Journal for Simulation and Multidisciplinary Design Optimization*.

### Peer-Reviewed Conference and Workshop Proceedings

2026 **Jain, R.** and Jacob, R. "Beyond Tool Execution: Evaluating Scientific MCP Interfaces with UXarray." *1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026*, Naples, Italy. Accepted; to be presented. Artifacts and data: doi:10.5281/zenodo.21463232.

2025 Gwinn, J., Wozniak, J. M., **Jain, R.**, Zhu, Y., Partin, A., Brettin, T., and Stevens, R. "A Workflow for Error Analysis for Drug Response Prediction via Statistical Standardization and Distribution Analysis." *Workshop on Workflows in Support of Large-Scale Science (WORKS)*.

2024 **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. "Enabling Data Reduction for Flash-X Simulations." *SC24 Workshops: DRBSD-10*.

2024 **Jain, R.**, Wozniak, J. M., Partin, A., Wilke, A., Zhu, Y., Vasanthakumari, P., Narykov, O., Overbeek, J., Weaver, R., Wang, C., Liu, Y., Weil, M. R., Brettin, T., and Stevens, R. "Cross-HPO: Optimizing Neural Networks for Cancer Drug Response Using Hyperparameter Tuning on Multiple Pharmacogenomic Datasets." *Computational Approaches for Cancer Workshop at SC24*.

2023 Wozniak, J. M., **Jain, R.**, Wilke, A., Weaver, R., Partin, A., Brettin, T., and Stevens, R. "An Automation Framework for Comparison of Cancer Response Models Across Configurations." *IEEE e-Science*.

2023 Dhruv, A., **Jain, R.**, O'Neal, J., Weide, K., and Dubey, A. "Framework and Methodology for Verification of a Complex Scientific Simulation Software, Flash-X." *Congress in Computer Science, Computer Engineering, and Applied Computing*.

2022 **Jain, R.**, Tang, H., Dhruv, A., Harris, J. A., and Byna, S. "Accelerating Flash-X Simulations with Asynchronous I/O." *PDSW*.

2021 **Jain, R.**, Shah, A., Mohd-Yusof, J., Wozniak, J. M., Brettin, T., Xia, F., and Stevens, R. "Probing Decision Boundaries in Cancer Data Using Noise Injection and Counterfactual Analysis." *CAFCW at SC21*.

2021 **Jain, R.**, Weide, K., Chawdhary, S., and Klostermann, T. "Checkpoint / Restart for Lagrangian Particle Mesh with AMR in Community Code FLASH-X." *SuperCheck*.

2015 **Jain, R.** and Tautges, T. J. "NEAMS MeshKit: Nuclear Reactor Mesh Generation Solutions." *International Congress on Advances in Nuclear Power Plants*.

2015 **Jain, R.**, Mahadevan, V., and O'Bara, R. "Simplifying Workflow for Reactor Assembly and Full-Core Modeling." *Mathematics and Computation (MC2015)*.

2014 **Jain, R.** and Tautges, T. J. "Generating Unstructured Nuclear Reactor Core Meshes in Parallel." *Procedia Engineering*.

2013 **Jain, R.** and Tautges, T. J. "PostBL: Post-Mesh Boundary Layer Mesh Generation Tool." *International Meshing Roundtable*.

2012 **Jain, R.** and Tautges, T. J. "RGG: Reactor Geometry (and Mesh) Generator." *International Congress on Advances in Nuclear Power Plants*.

2012 Mohanty, S., **Jain, R.**, Majumdar, S., Tautges, T. J., and Srinivasan, M. "Coupled Field Structural Analysis of HTGR Fuel Brick Using Abaqus." *International Congress on Advances in Nuclear Power Plants*.

2010 Tautges, T. J. and **Jain, R.** "Creating Geometry and Mesh Models for Nuclear Reactor Core Geometries Using a Lattice Hierarchy-Based Approach." *19th International Meshing Roundtable*. Best Paper Award.

### Conference Abstracts, Tutorials, and Presentations

2027 **Jain, R.** "Accurate Conservative Zonal Averaging for Unstructured Earth-System Grids: A Python Implementation of Robust Spherical Geometry Algorithms." *107th AMS Annual Meeting.* Accepted.

2026 **Jain, R.**, Abdulla, D., Jacob, R., Clyne, J., Eroglu, O., Medeiros, B., Chen, H., and Ullrich, P. "A Model Context Protocol Server for Agentic Analysis of Unstructured Earth-System Meshes." Presented at *SciFoundationModels 2026 (SciFM26).*

2026 Eroglu, O., Chen, H., Chmielowiec, P., Clyne, J., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. M. "Latest Developments in UXarray: A Python Package Supporting Kilometer-Scale Model Outputs in Native Unstructured Grids." *106th AMS Annual Meeting.*

2026 Clyne, J., Eroglu, O., Jacob, R., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. "UXarray: A Python Package for the Analysis of Kilometer-Scale Atmosphere and Ocean Model Outputs." *EGU General Assembly (EGU26).*

2025 Eroglu, O., Chen, H., Chmielowiec, P., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. M. "UXarray: Extending Xarray to Support the Analysis of Native, Kilometer-Scale Unstructured Grids in Python." *105th AMS Annual Meeting.*

2025 Clyne, J., Chen, H., Chmielowiec, P., Eroglu, O., Hannay, C., Jacob, R., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. "UXarray: Extending Xarray for Enhanced Support of Unstructured Grids." *EGU General Assembly.*

2024 **Jain, R.** "Tutorial: UXarray for Analysis of Unstructured Climate Data." *SC24.*

2024 Chmielowiec, P., Chen, H., DeCiampa, C., Eroglu, O., Hannay, C., Jacob, R. L., **Jain, R.**, Loft, R., Medeiros, B., Sun, L., Ullrich, P., and Zarzycki, C. M. "UXarray: Extending Xarray with Support for Unstructured Grids." *104th AMS Annual Meeting.*

2024 Eroglu, O., Chen, H., Chmielowiec, P., Clyne, J., DeCiampa, C., Hannay, C., Jacob, R., **Jain, R.**, Loft, R., Medeiros, B., Sun, L., Ullrich, P., and Zarzycki, C. "UXarray: Extensions to Xarray to Support Unstructured Grids." *EGU General Assembly.*

2024 Clyne, J., Chen, H., Chmielowiec, P., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P. A., and Zarzycki, C. M. "UXarray: An Extension to Xarray Supporting Analysis and Visualization of Kilometer-Scale Model Outputs." *AGU Fall Meeting.*

2024 **Jain, R.** "UXarray Tutorial." *ESDS Annual Event.* Video.

2023 **Jain, R.** "UXarray for Unstructured Climate Data." *SciPy 2023.*

2023 **Jain, R.** "Data Reduction for FLASH-X Simulations." *HDF User Group 2023.*

2023 Clyne, J., Chen, H., Chmielowiec, P., DeCiampa, C., Eroglu, O., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Sun, L., Ullrich, P. A., and Zarzycki, C. M. "UXarray: A Community-Developed Tool for the Analysis and Visualization of Model Output on Unstructured Grids in Python." *AGU Fall Meeting.*

2023 Jacob, R. L., Ullrich, P. A., Zhang, C., **Jain, R.**, Chen, H., Po-Chedley, S., and Vo, T. "Creating New Standards to Simplify Earth System Model Analysis." *AGU Fall Meeting.*

2019 Sever, G., Obabko, A., Jacob, R., **Jain, R.**, and Catlett, C. "Simulations of Idealized and Real Urban Boundary Layer Environments Using NWP and CFD Models." *18th Conference on Mesoscale Processes.*

2019 Jacob, R. L., Sever, G., Obabko, A., **Jain, R.**, and Catlett, C. E. "CFD Simulations of Flow in Realistic Urban Geometries Initialized from Weather Models." *AGU Fall Meeting.*

2018 Obabko, A., **Jain, R.**, Jacob, R., Paoli, R., Sever, G., Grindeanu, I., and Fischer, P. "Nek5000 Wall-Resolved LES for Urban Geometries." *10th International Conference on Urban Climate / 14th Symposium on the Urban Environment.*

2018 Silva, M. P., Sharma, A., Budhathoki, M., **Jain, R.**, and Catlett, C. E. "Neighborhood Scale Heat Mitigation Strategies Using Array of Things Data in Chicago." *AGU Fall Meeting.*

2018 Sever, G., Jacob, R., and **Jain, R.** "Mesoscale to Microscale Weather Simulations over the Chicago Downtown Region." *10th International Conference on Urban Climate / 14th Symposium on the Urban Environment.*

## Technical Reports, Manuals, and Other Publications

2020 Goldring, N., Bruhwiler, D., Nash, B., Wu, Z., Nagler, R., Carter, J., Lerch, J., Suthar, K., Den Hartog, P., **Jain, R.**, and Mahadevan, V. "Multiphysics Design and Optimization of Complex Vacuum Chambers (Phase II Final Technical Report)."

2017 **Jain, R.** "Reactor Geometry (&Mesh) Generator Users Manual."

2016 **Jain, R.**, Vanderzee, E., Grindeanu, I., and Mahadevan, V. "Mesh Generation and Algorithm Development for NEAMS."

2015 **Jain, R.**, Vanderzee, E., and Mahadevan, V. "Update on Development of Mesh Generation Algorithms in MeshKit." ANL/MCS-TM-355.

2015 **Jain, R.** and Mahadevan, V. "Documentation for MeshKit-Reactor Geometry (&mesh) Generator." ANL/MCS-TM-354.

2015 Mahadevan, V., Grindeanu, I. R., Ray, N., **Jain, R.**, and Wu, D. "SIGMA Release v1.2: Capabilities, Enhancements and Fixes." ANL/MCS-TM-357.

2015 Merzari, E., Shemon, E. R., Yu, Y., Thomas, J. W., Obabko, A., **Jain, R.**, Mahadevan, V., Solberg, J., Ferencz, R., and Whitesides, R. "Full Core Multiphysics Simulation with Offline Mesh Deformation." ANL/NE-15/42.

2015 Merzari, E., Shemon, E. R., Yu, Y. Q., Thomas, J. W., Obabko, A., **Jain, R.**, Mahadevan, V., Tautges, T., Solberg, J., Ferencz, R. M., and Whitesides, R. "Multi-physics Demonstration Problem with the SHARP Reactor Simulation Toolkit." ANL/NE-15/44.

2014 **Jain, R.** "Report on FY11 Extensions to MeshKit and RGG." ANL/MCS-TM-316.

2014 **Jain, R.** and Mahadevan, V. "2014 MeshKit Release." ANL/MCS-TM-344.

2014 Tautges, T. J. and **Jain, R.** "Mesh Copy/Move/Merge Tool for Reactor Simulation Applications." ANL/MCS-TM-343.

2013 Tautges, T. J., Fischer, P., Grindeanu, I., **Jain, R.**, Mahadevan, V., Obabko, A., Smith, M. A., Merzari, E., and Ferencz, R. M. "SHARP Assembly-Scale Multiphysics Demonstration Simulations." ANL/NE-13/9.

2013 **Jain, R.** and Tautges, T. J. "MeshKit." ANL/MCS-TM-336.

2012 Tautges, T. J., Fischer, P., Grindeanu, I., **Jain, R.**, Mahadevan, V., Obabko, A., Smith, M. A., Hamilton, S., Clarno, K., and Baird, M. "A Coupled Thermal / Hydraulics – Neutronics – Fuel Performance Analysis of an SFR Fuel Assembly." Report to the U.S. Department of Energy.

## Software and Open Source

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|---|---|
| **UXarray** | Lead developer; Python library for unstructured climate grid analysis used by NCAR and DOE labs. GitHub ů Docs. |
| **UXarray MCP** | MCP server exposing UXarray as typed AI-agent tools with Globus Compute HPC backend; structured provenance on every call. Presented at SciFM26. |
| **FLASH-X** | I/O and compression lead for a million-line multiphysics simulation engine (astrophysics, combustion, fluid dynamics). R&D 100 Award, 2022. |
| **CANDLE / IMPROVE** | HPO workflow (Swift/T, mlrMBO, DEAP, Hyperopt) and `improvelib` benchmark framework for cancer drug response; GitHub Actions CI/CD and UNO dual-branch model. R&D 100 Award, 2023. |
| **SFNO** | PyTorch climate emulator ported to Aurora's Intel XPU stack (60,000+ GPUs): PMIX/PALS launcher, XPU/CUDA AMP, first stable DDP baseline, measured one-node 12-rank run. |
| **MeshKit** | Principal investigator; open-source C++ reactor mesh generation toolkit. Parallel CoreGen: 101M-element MONJU mesh on 712 processors. Source. |

## Professional Service

- Program committee member, 1st International Workshop on Agentic AI for HPC (AgenticAI4HPC) at SC26, Chicago, IL.
- Program committee member, 1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026, Naples, Italy.
- SBIR / STTR proposal reviewer for the U.S. Department of Energy.
- Panelist, "Revolutionizing Public Infrastructure: The Impact of AI and Machine Learning," 5th Infraday Midwest Event.
- Reviewer for the *Journal of Open Research Software* and NumGrid.
- Program committee member, NumGrid 2020.
- Session chair, Computational Geometries session, MC2015.
- Volunteer, South Side Science Festival, University of Chicago (2023, 2024).

## Professional Memberships

- Association for Computing Machinery (ACM), since 2018.
- American Meteorological Society (AMS).
- American Nuclear Society (ANS), 2011 – 2015.

## Mentorship

- Mentored students, research aides, and staff on scientific Python, HPC workflows, systems engineering, and reproducible software development.
- Rylie Weaver (Research Aide, 2022 – 2024): IMPROVE project, HPO techniques for cancer drug response prediction.
- Aaron Zedwick (Student, 2023 – 2024): UXarray development, dual mesh routines, and remapping functionality.
- Mark Bartoszek (Systems Administration, 2023): mentoring on systems administration practices at Argonne.
- Brett Rhodes (SULI Summer Student, 2013): MeshKit documentation and examples.
- Dayan Abdulla (Student, 2024 – present): UXarray MCP server development, AI-agent dataset exploration, and Globus Compute integration for HPC execution.